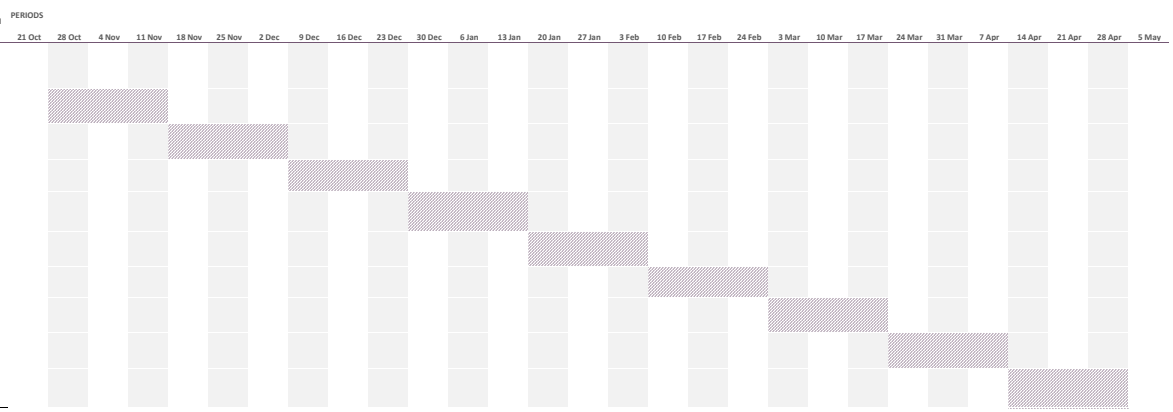
Project Planner

ACTIVITY	PLAN START	PLAN DURATION
Week 1–2 (14–27 Oct 2025): Project Setup & Ethics Approval — Define project scope, research goals, and data requirement. Complete ethical review and confirm dataset licensing compliance. Milestone: Project plan and ethics approved.	25 Oct 2025	3
Week 3–5 (28 Oct – 17 Nov 2025): Dataset Acquisition & Preparation — Gather open-source fish image datasets, verify data quality, clean and label images, and organize data for model training. Milestone: Dataset ready for AI model training.	28 Oct 2025	21
Week 6–8 (18 Nov – 8 Dec 2025): Baseline AI Model Development — Develop and train a baseline convolutional neural network for species recognition using cleaned datasets. Milestone: Working prototype with initial accuracy results.	##########	21
Week 9–11 (9–29 Dec 2025): Fish Health Detection Module — Enhance the AI model to detect visual indicators of fish illness and stress. Begin testing under varied lighting and image conditions. Milestone: Health detection accuracy validated on test data.	9 Dec 2025	21
Week 12–14 (30 Dec 2025 – 19 Jan 2026): Invasive Species Identification — Train and test the model to identify invasive or non-native species and compare results across datasets. Milestone: Invasive species detection functioning reliably.	30 Dec 2025	21
Week 15–17 (20 Jan – 9 Feb 2026): Population Tracking and Counting System — Implement algorithms for counting fish and detecting changes in population size across images and videos. Milestone: Population tracking system integrated.	20 Jan 2026	21
Week 18–20 (10 Feb – 2 Mar 2026): Evaluation and Optimization — Conduct performance testing, evaluate accuracy, handle bias detection, and refine model parameters. Milestone: Model performance optimized and benchmarked.	10 Feb 2026	21
Week 21–23 (3–23 Mar 2026): Dashboard and Visualization Development — Design a simple dashboard to display data on fish health, species distribution, and population trends. Milestone: Dashboard prototype complete.	3 Mar 2026	21
Week 24–26 (24 Mar – 13 Apr 2026): Integration and Final Testing — Integrate all system components, conduct user testing, and document software reliability. Milestone: Fully functional SmartAquatic system ready for final report.	##########	21
Week 27–29 (14 Apr – 5 May 2026): Final Report and Presentation — Prepare final documentation, evaluation summary, and project presentation materials. Milestone: SmartAquatic project submitted successfully.	14 Apr 2026	21